

Week 3 Report — System Design, Architecture & Proof of Concept

Programme: IIMSTC – IoT & Embedded Systems (Cohort 4)

Team Size: 15 Members

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Day 1 — Methodology Planning

The team defined a structured methodology for implementing the IoT-driven cold chain system. The approach included data acquisition using sensors, real-time data transmission through microcontrollers, cloud-based monitoring, and alert generation. A step-by-step workflow was established to ensure smooth progression from hardware setup to cloud integration and system testing.

Day 2 — System Design & Architecture

A detailed system architecture was designed, outlining the interaction between sensors (DHT22 and reed sensors), microcontroller (ESP32), cloud platform, and user interface dashboard. The design consisted of multiple layers including sensor layer for environmental monitoring, processing layer using ESP32 for data handling, communication layer using Wi-Fi protocols (MQTT/HTTP), and cloud/dashboard layer for data visualization and alerts. This structured architecture ensured clarity, scalability, and efficient implementation.

Day 3 — Simulation of Sensors (Wokwi Platform)

The team conducted simulations using the Wokwi platform to validate system behavior before hardware implementation. The DHT22 sensor was simulated to obtain temperature and humidity readings, while the reed sensor was used to detect door open/close status. Serial output was verified to ensure accurate data acquisition. This step helped identify potential issues early and refine the system logic.

Day 4 — Dashboard Development

A cloud-based dashboard was developed to visualize real-time data. The dashboard displayed temperature values, humidity levels, and door status (open/closed). The interface was designed for clarity and ease of monitoring, enabling users to efficiently track cold chain conditions in real time.

Day 5 — Alert System Integration (Blynk Platform)

An alert mechanism was implemented using the Blynk platform. Alerts were triggered when temperature exceeded the defined threshold range (2°C–8°C), when abnormal humidity levels were detected, and when unauthorized door opening occurred. Real-time notifications ensured immediate response to potential risks, improving overall system reliability.

Day 6 — Proof of Concept (PoC) Validation

The team tested the integrated system under multiple scenarios to validate functionality. This included continuous sensor data reading and transmission, real-time dashboard updates, and alert triggering when threshold limits were breached. The successful testing confirmed the working capability of the proposed system and validated the Proof of Concept.

Summary

Week 3 focused on transforming the finalized problem into a practical system through structured methodology, detailed design, and simulation. The team successfully developed the system architecture, validated sensor behavior through simulation, and implemented a functional dashboard with alert mechanisms. The completion of the Proof of Concept established a strong technical foundation for further development and full-scale implementation in the next phase.